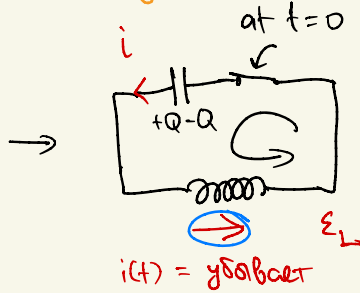
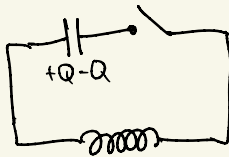


Resnick Chapter 31: AC Circuits

• LC Circuits

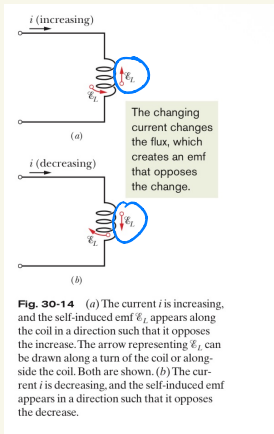
Здесь конденсатор работает как батарея



$$+ \frac{Q}{C} + L \frac{di}{dt} = 0$$

Ex (case study)

Закон Ленца: induced voltage tries to maintain initial state



- if $i = \text{decreasing}$, ϵ_L (induced) will be in i -direction to help restore the decreasing amount.
- if $i = \text{increasing}$, ϵ_L (induced) will be in opposite direction

$$\frac{q(t)}{C} + L \frac{di}{dt} = 0$$

$$i = \frac{dq}{dt}$$

$$\frac{q}{C} + L \frac{d^2 q}{dt^2} = 0$$

$$\frac{di}{dt} = \frac{d}{dt} \left(\frac{dq}{dt} \right) = \frac{d^2 q}{dt^2}$$

$$\boxed{\frac{d^2 q}{dt^2} + \frac{1}{LC} q(t) = 0}$$

Formula of SHO

$$\boxed{\ddot{x} + \omega^2 x = 0}$$

$$\frac{d^2 q}{dt^2} + \frac{1}{LC} q(t) = 0$$

Formula of SHO

$$\ddot{x} + \omega^2 x = 0$$

• частота колебания

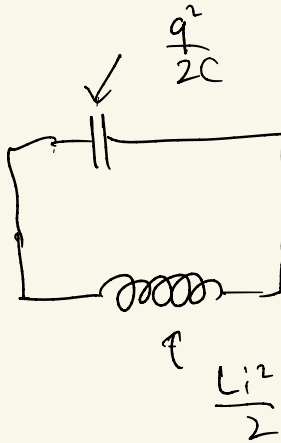
$$\omega^2 = \frac{1}{LC}$$

$$q(t) = Q_0 \sin(\omega t + \phi)$$

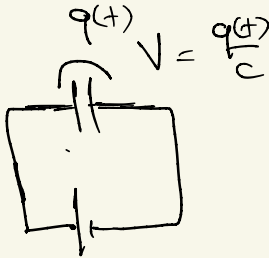
w/ general solution

$$x(t) = x_0 \sin(\omega t + \phi)$$

$$x(t) = A \sin \omega t + B \cos \omega t$$



energy stored in magnetic field



Energy of a capacitor = work done to charge it up to $0 \rightarrow Q$.

Необходимая работа при увеличении dq к конденсатору
= запасом $q(t)$: $dW = (dq) V_C = (dq) \frac{q}{C}$

$$W_{\text{total}} (\text{to charge } C (0 \rightarrow Q)) = \int dW$$

$$W_{\text{total}} = \int_0^Q \frac{q}{C} dq$$

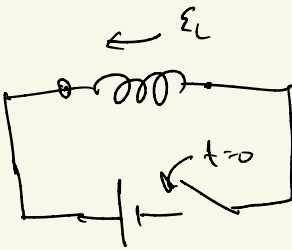
$$= \frac{q^2}{2C} \Big|_0^Q$$

$$\boxed{= \frac{Q^2}{2C}}$$

Energy stored in capacitor with charge Q .

$\mathcal{E}$ - is stored in electric field.

• Energy stored in inductor:



$$\text{Power} = P = iV$$

$$= iL \frac{di}{dt}$$

$$P = \frac{dW}{dt} \Rightarrow dW = P dt$$

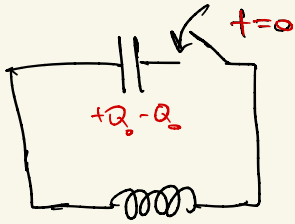
$$\text{Total work} = \int_0^t P dt$$

$$E_L = W_{\text{total}} = \int_0^t P dt = \int_0^I Li di = \frac{LI^2}{2} \Big|_0^I$$

$$\boxed{E_L = \frac{LI^2}{2}}$$

Energy stored in an inductor, energy is stored in B-field.

Go back to LC - circuit



$$Q(t) = Q_0 \sin(\omega t + \phi)$$

At $t=0$, $Q(t=0) = Q_0$

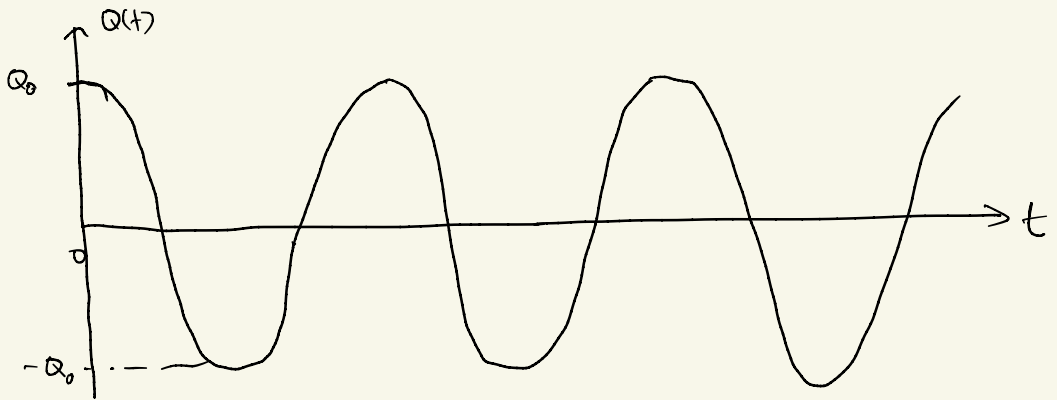
$$Q(t=0) = Q_0 \sin(\underbrace{\omega t + \phi}_{=1})$$

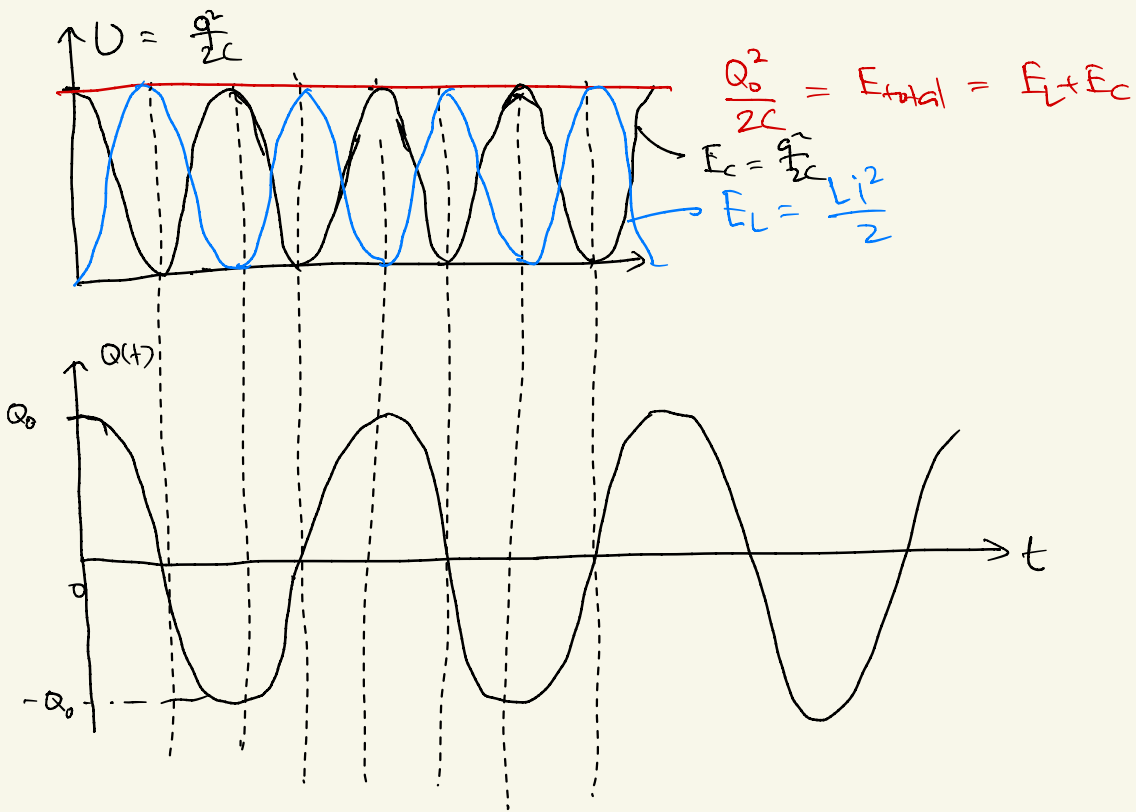
$$\boxed{\phi = \frac{\pi}{2}}$$

$$\boxed{Q(t) = Q_0 \sin(\omega t + \frac{\pi}{2})}$$

$\Downarrow$

$$\boxed{Q(t) = Q_0 \cos(\omega t)}$$





$$E_L = \frac{Li^2}{2}$$

$$i = \frac{dq}{dt} = \frac{d}{dt} \left(Q_0 \sin(\omega t + \frac{\pi}{2}) \right)$$

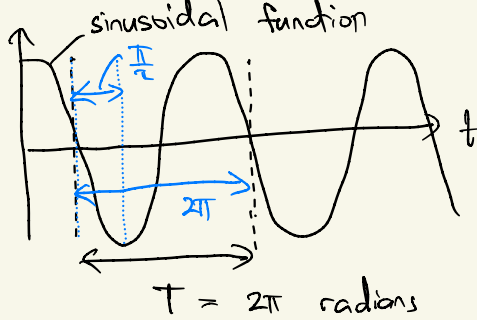
$$= \omega Q_0 \cos(\omega t + \frac{\pi}{2})$$

Here: $\frac{d}{dt} \left[\sin(\omega t + \frac{\pi}{2}) \right] = \frac{d(\sin u)}{du} \frac{du}{dt}$ chain rule

$$= \cos u \frac{du}{dt} = \omega \cos(\omega t + \frac{\pi}{2})$$

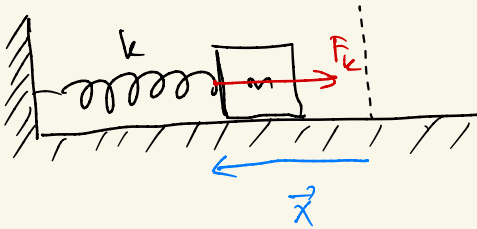
$$\cos \theta = \sin(\theta + \frac{\pi}{2})$$

there's a $\frac{\pi}{2}$ phase diff. btwn q and i (U_c and U_L)



$\frac{\pi}{2}$ is $\frac{1}{4}$ of total period

Recall mass-spring system:



$$\vec{F}_k = -k\vec{x}$$

$$F = ma$$

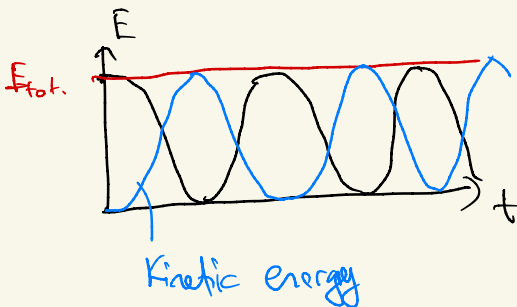
$$-kx = m \frac{dv}{dt} = m \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$x(t) = x_0 \sin(\omega t + \phi)$$

$$E_{\text{tot}} = \frac{kx^2}{2} + \frac{mv^2}{2}$$

$$= \frac{kx_{\text{max}}^2}{2}$$



Kinetic energy $KE = \frac{1}{2}mv^2$

$$v = \frac{dx}{dt} = \frac{d}{dt} [x_0 \sin(\omega t + \phi)]$$

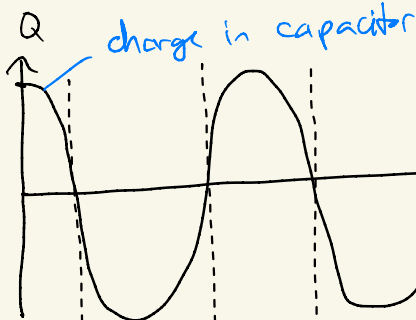
$$= \omega x_0 \cos(\omega t + \phi)$$

$$U_k = \frac{1}{2}kx^2 = \frac{1}{2}kx_0^2 \underbrace{\sin^2(\omega t + \phi)}$$

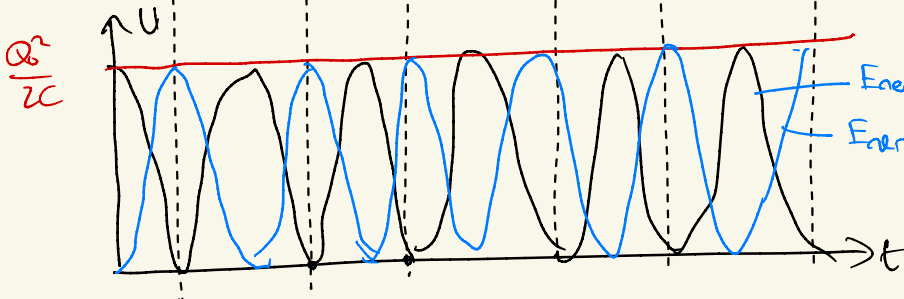
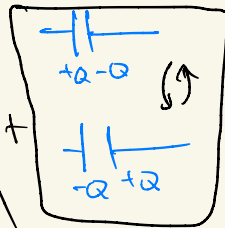
$$U_{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2x_0^2 \underbrace{\cos^2(\omega t + \phi)}$$

$\frac{\pi}{2}$ phase diff.

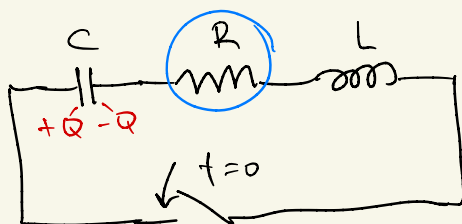
identical in form to LC oscillations!
math is same



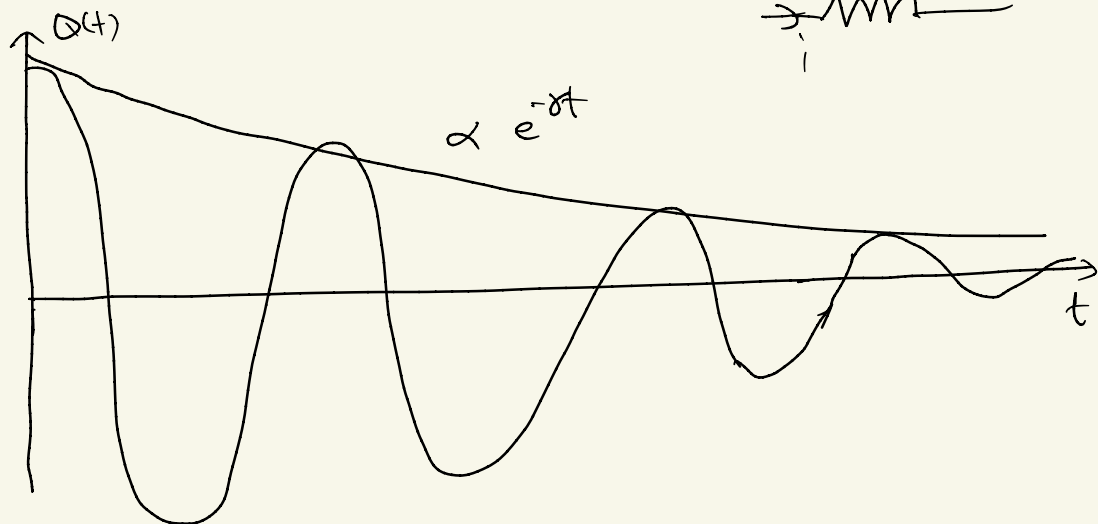
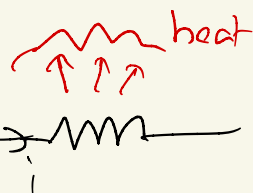
$+Q \rightarrow -Q$
 is just polarity

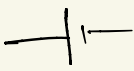


Q: What happens if you add R?



R will dissipate energy to heat

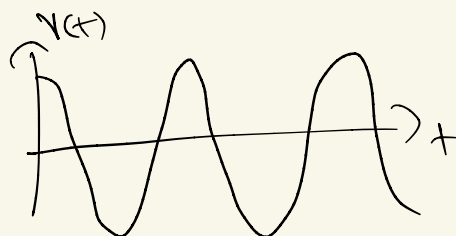


in AC circuits instead of  DC battery

You will have

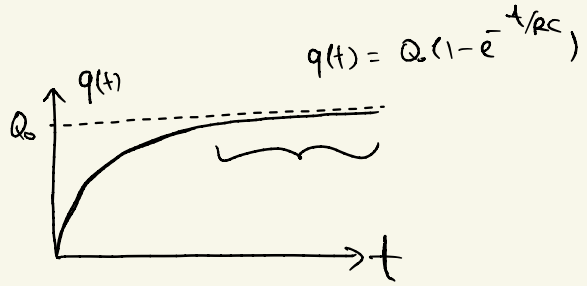
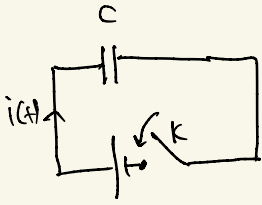
 alternating voltage source

$$V_s \sin(\omega t + \phi)$$

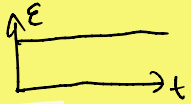


AC Circuits

DC picture

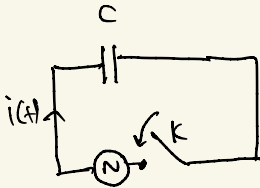


Up to this time, we were looking at this.



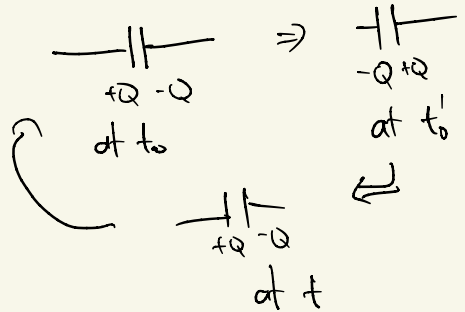
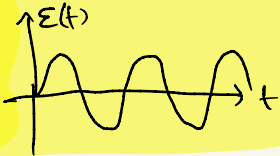
$$i(t) = \frac{dq}{dt}$$

AC Picture: when we have time-varying voltage source



$$V_0 \sin(\omega t)$$

capacitor will charge and discharge continuously



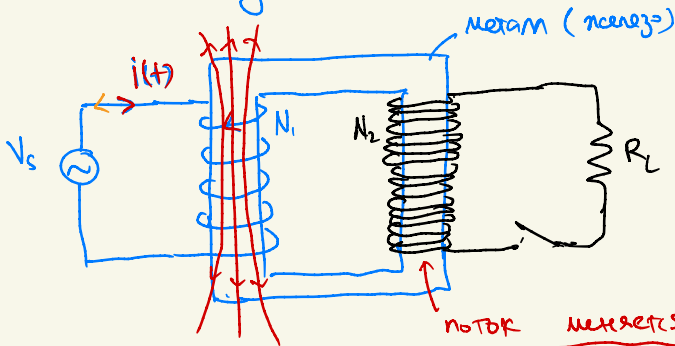
In general, in AC circuits you have $\epsilon(t) = \epsilon_m \sin(\omega t + \phi)$ time-varying voltage source.

Q: Why do we need AC voltage source (i.e., why to complicate things)?

A: In practice all of our sockets at home operate at AC, not in DC mode.

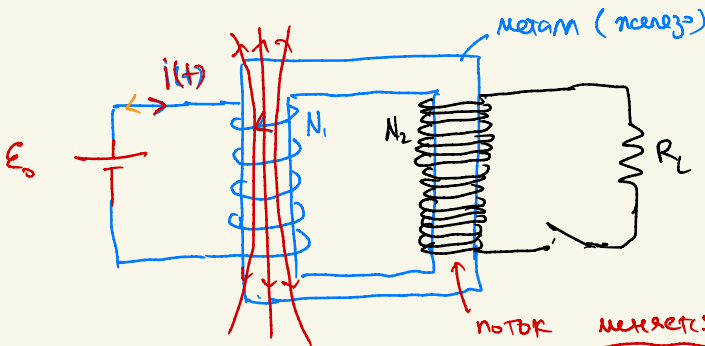
↓
быстрее и проще

Secret lies in Transformers: работает по закону Фарадея, поэтому AC is handy there.



$$\mathcal{E}_{ind} = - \frac{d\Phi_B}{dt}$$

$$B \propto i(t)$$



поток меняется во времени

$$\mathcal{E}_{ind} = - \frac{d\Phi_B}{dt} = 0$$

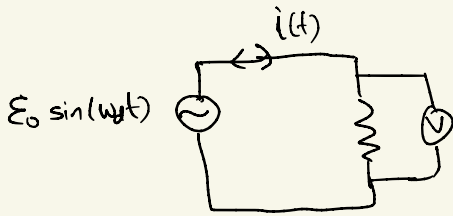
если $V_s = \underline{DC}$

$$\frac{d\Phi_B}{dt} = 0$$

потому что здесь $i(t) = \text{const.}$

Phasors

① Resistors in AC source:

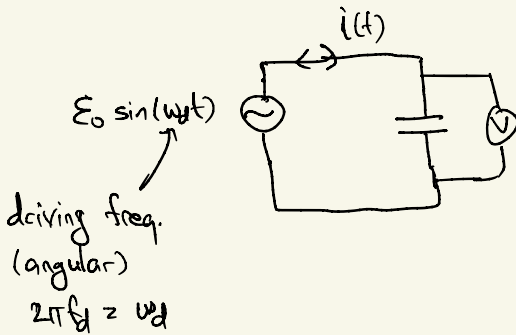


$$V_R = iR$$

$$i(t) = \frac{V_R}{R} = \frac{\mathcal{E}_0 \sin(\omega_d t)}{R}$$

$$V(t) = \mathcal{E}_0 \sin(\omega_d t)$$

② Capacitors in AC source:



$$V_C = \frac{q(t)}{C} \rightarrow q(t) = CV_C$$

$$i(t) = \frac{dq}{dt} = C \frac{dV}{dt}$$

$$i(t) = C \frac{d(\mathcal{E}_0 \sin(\omega_d t))}{dt}$$

$$= \omega_d C [\mathcal{E}_0 \cos(\omega_d t)]$$

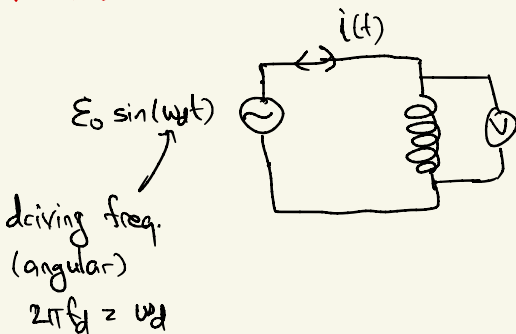
Key point: Voltage in C lags 90° behind current

$$i_C(t) = \omega_d C \mathcal{E}_0 \sin(\omega_d t + \frac{\pi}{2})$$

$$V(t) = \mathcal{E}_0 \sin(\omega_d t)$$

phase difference

③ Inductors in AC source:



$$\mathcal{E}_L = L \frac{di}{dt} = \mathcal{E}_0 \sin(\omega_d t)$$

$$i(t) = \int_0^t di = \int_0^t \frac{\mathcal{E}_0}{L} \sin(\omega_d t) dt$$

$$i(t) = -\cos(\omega_d t) \frac{\mathcal{E}_0}{\omega_d L}$$

$$i(t) = -\cos(\omega t) \frac{\mathcal{E}_0}{\omega L}$$

$$i(t) = \frac{\mathcal{E}_0}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

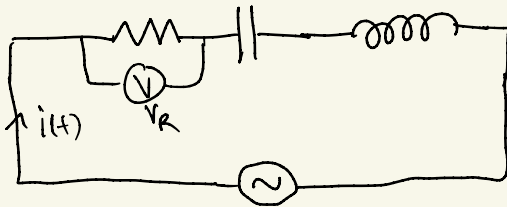
$$V(t) = \mathcal{E}_0 \sin(\omega t)$$

phase difference

Key point: Voltage in L leads 90° behind current

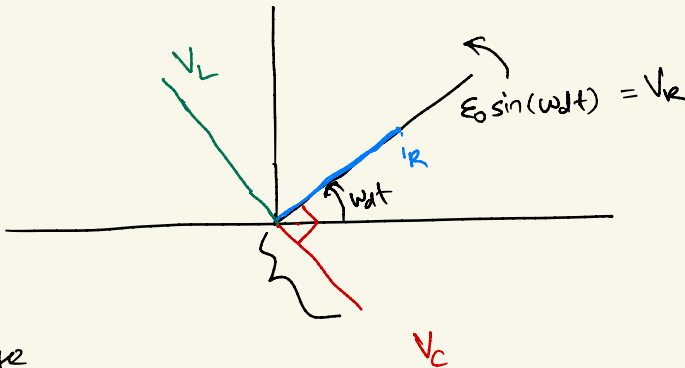
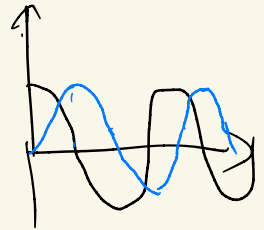
Далее рассмотрим резонанс в цепи переменного тока: RLC circuit

Key point: зная по всем элементам прохождение тока и тот же ток $i(t)$!



$$\mathcal{E}_0 \sin(\omega t)$$

$$\mathcal{E}_0 \cos(\omega t + \phi)$$



Total Voltage

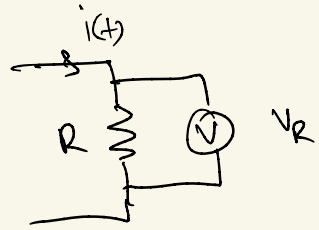
$$V_{\text{tot.}} = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Let's dig more through current going through R, C, and L

• R

$$i_c(t) = \frac{V_R}{R} \rightarrow V_R = i_c R$$

Ohm's law: $V = IR \rightarrow \boxed{R}$

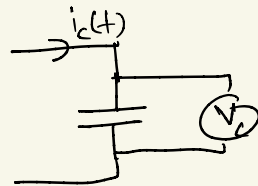


• C

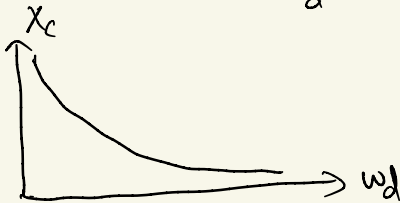
$$i_c(t) = \omega_d C V_c$$

Ohm's law: $V = IX_c$

$$V_c = \frac{i_c}{\omega_d C} \rightarrow \boxed{X_c = \frac{1}{\omega_d C}}$$



definition: capacitive reactance
(aka, capacitance)

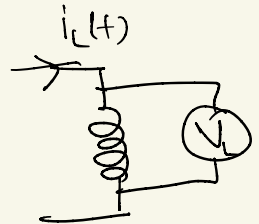


• L

$$i_L = \frac{V_L}{\omega_d L} \rightarrow V_L = i_L \omega_d L$$

Ohm's law: $V = IX_L$

$$\boxed{X_L = \omega_d L}$$



$$\varepsilon_{ind} = - \frac{d\Phi}{dt} \propto \frac{di}{dt}$$

if current doesn't change $\rightarrow$ no induction
and vice versa

Definition: impedance Z (aka: equivalent resistance)

$$V_{tot} = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V_R = IR$$

$$V_L = IX_L$$

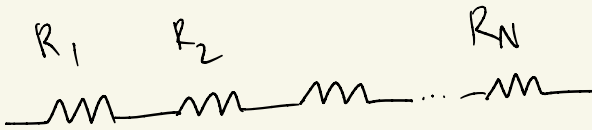
$$V_C = IX_C$$

$$V_{tot} = I \sqrt{R^2 + (X_L - X_C)^2}$$

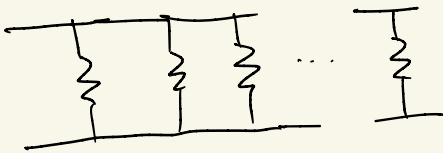
$$Z = \frac{V_{tot}}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

= 0 in resonance

Let's compare with circuits (series) & parallelных тех:



$$R_{eq} = \sum_{i=1}^N R_i$$



$$\frac{1}{R_{eq}} = \sum_{i=1}^N \frac{1}{R_i}$$

In capacitors it's inverse

Но как вы уже мы работаем как со скалярными величинами! А в AC, $R \perp X_L$ и X_C ! Мы не можем линейно их сложить; мы сложим их как векторы (по теореме Пифагора)

AC Circuits are very important in Electronics and Olympiads as well.

APHO 2025 Exp

1 of 19

PHYSICS OF INDUCTION COOKING

Solution of the Experimental Problem
Asian Physics Olympiad, Dhahran Saudi Arabia 2025
Oki Gunawan, Herry Kwee, Hendra Kwee, Yendi, M. Morteza Mudrick
v.3.0 2025/5/9

1. EXP#1: CHARACTERIZATION OF THE INDUCTION COIL

1.1 Series RLC circuit

To determine L using resonance experiment we setup a series R - L - C circuit as shown below. We measure the source voltage from the Function Generator (FG): V_S and measure the current I by measuring the voltage across "shunt resistance" $R=1\ \Omega$. $I = V_S / R$.

We use the digital oscilloscope to measure the voltage. For convenient and quick measurement, we can fix one probe terminal (e.g.) on node #1 and repetitively measure voltage on node#2 and node#3 to measure source voltage V_S and V_R respectively.

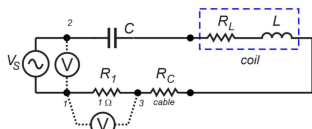


Figure 1. The series RLC circuit

The cables involved in the circuit are two pieces of banana-to-pins cable (item#6). Using ohmmeter we obtain: $R_C = 0.09\ \Omega$.

1.2 Resonance series RLC circuit

Then we determine the resonant condition where the impedance $Z = V_S / I$ reaches minimum or conductance $G = I / V_S$ reaches maximum. A smart student should quickly scan the frequency first to quickly find the resonant condition and suitable frequency range before taking data, e.g. fix V_S and scan f as a function of frequency. We note that V_S could vary due changing load impedance. The data is shown below:

1

SOLUTION

10:33 AM Sun Aug 24

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2 of 19

Units	C(X)	R(X)	V(S)(Y)	V(R)(Y)	G(Y)
Units	uF	Hz	V	V	S
Units			(Vrms)	(Vrms)	(Vrms)
1	0.47	16.3	11.987	0.748	0.08223
2		21.6	9.133	0.887	0.10586
3		23.9	4.379	0.816	0.14077
4		28.4	1.485	0.373	0.23481
5		31.6	0.799	0.327	0.40026
6		32.5	0.742	0.32	0.43127
7		33	2.353	1.187	0.52318
8		33.8	2.283	1.187	0.50679
9		35.3	0.885	0.335	0.48895
10		36.6	2.435	1.149	0.47187
11		36.7	1.351	0.544	0.40296
12		37.5	0.884	0.32	0.38784
13		38.8	1.047	0.316	0.30181
14		40.5	1.332	0.335	0.2515
15		43.2	1.922	0.38	0.19771
16		48	2.335	0.484	0.14343
17		54.1	1.915	0.622	0.10389
18		64.8	4.719	0.373	0.07904

Table 1. RLC resonance data for $C=470\ \text{nF}$ and $C=2200\ \text{nF}$

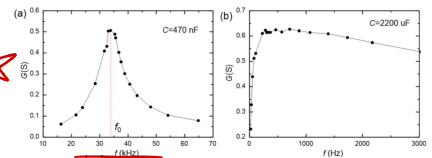


Figure 2. Resonance conductance plot of the RLC circuit with: (a) $C = 470\ \text{nF}$; (b) $C = 2200\ \text{nF}$.

Units	C(X)	R(X)	L(Y)
Units	uF	Hz	uH
mmnts		resonance	
1	0.47	33800	47.17
2	2200	400	71.96

Table 2. Results of L determination from RLC resonance

2

SOLUTION

The results are shown in Table 2. The resonance frequency is given as: $\omega = 1 / \sqrt{LC}$, thus

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5 of 19

Measurement of R_2 directly with the multimeter for verification is also acceptable, we obtain:
 $R_2 = (0.47 \pm 0.03)\ \Omega$.

2. EXP#2: MUTUAL INDUCTION AND SKIN DEPTH

2.1 Mutual Induction

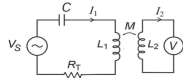


Figure 4. Mutual induction setup

2.2 Mutual inductance experiment

Here we perform the measurement twice where the primary is coil#1 and secondary is coil#2 and then we swap them.

$$V_1 = -M \frac{dI_2}{dt} + L_1 \frac{dI_1}{dt}, \quad V_2 = -M \frac{dI_1}{dt} + L_2 \frac{dI_2}{dt} \quad (6)$$

$$Z' = \frac{V_1}{I_1} = -j\omega M \quad (7)$$

The self inductance contribution is negligible since the second coil is connected to voltmeter and $I_2 \approx 0$. Then we can tabulate the "impedance" $Z = V_2 / I_1$ as a function of frequency and extract M .

Units	f(X)	I1(Y)	V2(Y)	Z(Y)		f(X)	I2(Y)	V1(Y)	Z(Y)
Units	Hz	A	V	Ohm	Units	Hz	A	V	Ohm
1	1020	3.539	0.098	0.02713	1	1010	3.52	0.136	0.03864
2	1860	3.082	0.184	0.05987	2	2000	3.006	0.2	0.06653
3	3000	2.888	0.241	0.08312	3	3000	2.512	0.27	0.10748
4	4640	2.131	0.278	0.13046	4	4000	2.112	0.269	0.14157
5	8010	1.827	0.295	0.16147	5	5000	1.808	0.316	0.17478
6	8000	1.617	0.312	0.19285	6	6000	1.56	0.337	0.21603
7	7010	1.427	0.32	0.22426	7	7020	1.355	0.339	0.25019
8	8860	1.286	0.342	0.27229	8	8660	1.21	0.358	0.29587
9	8140	1.104	0.38	0.31703	9	9010	1.104	0.369	0.33424
10	10100	1.012	0.368	0.36375	10	10200	0.997	0.373	0.37412

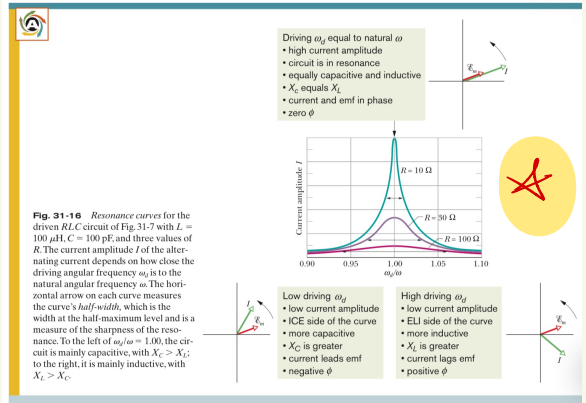
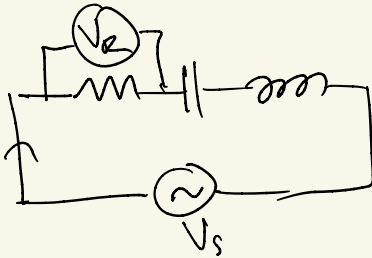
Table 5. Mutual inductance determination

5

SOLUTION

Was all about AC Circuits and Mutual Inductance

Notes:

ResonancePower delivered through $R = IV_R$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

in resonance :

$$\underline{X_L = X_C}$$

 $\hookrightarrow$ in resonance $= 0$

$$\omega_d L = \frac{1}{\omega_d C}$$

$$\omega_d^2 = \frac{1}{LC}$$

$$\boxed{\omega_d = \frac{1}{\sqrt{LC}}}$$

resonance condition

$$Z \text{ reduces to } \underline{Z = \sqrt{R^2 + 0^2} = R}$$

Transformers

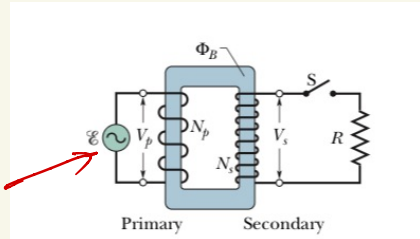


Fig. 31-18 An ideal transformer (two coils wound on an iron core) in a basic transformer circuit. An ac generator produces current in the coil at the left (the *primary*). The coil at the right (the *secondary*) is connected to the resistive load R when switch S is closed.

$$V_p = N_p \mathcal{E}$$

$$V_s = N_s \mathcal{E}$$

$$-\frac{d\Phi_B}{dt} \propto \frac{di}{dt} = \frac{1}{R} \frac{d\mathcal{E}}{dt} \downarrow L \frac{di}{dt}$$

$$\frac{V_p}{N_p} = \frac{V_s}{N_s}$$

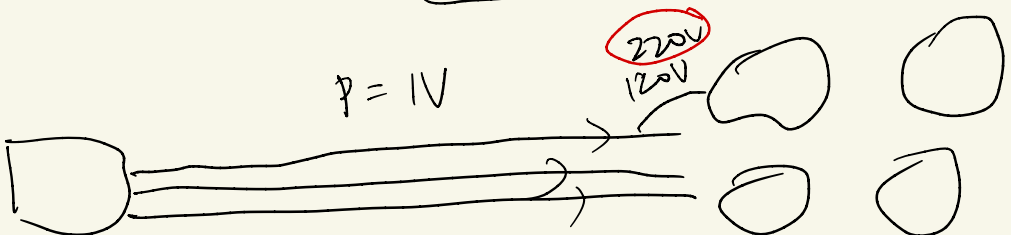
$$V_s = V_p \frac{N_s}{N_p}$$

Case 1: $N_s > N_p$

$$V_{\text{secondary}} > V_{\text{primary}}$$

Case 2: $N_s < N_p$

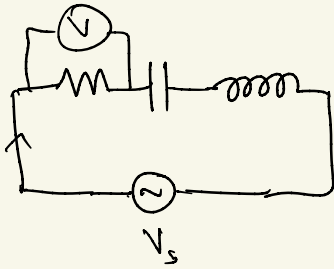
$$V_{\text{sec}} < V_{\text{primary}}$$



Note 2: Series RLC

vs

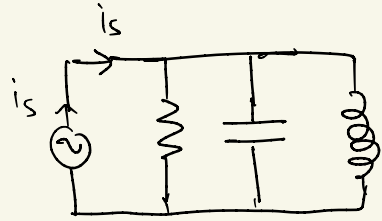
Parallel RLC



$$i_s = i_R = i_C = i_L$$

$$V_s = V_R + V_C + V_L$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$



$$i_s = i_R + i_C + i_L$$

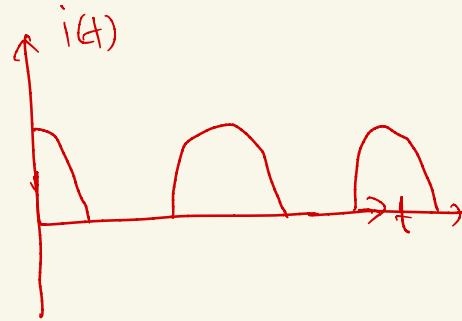
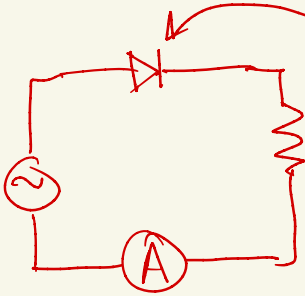
$$V_s = V_R = V_C = V_L$$

$$\frac{1}{Z} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L - X_C}\right)^2}$$

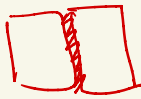
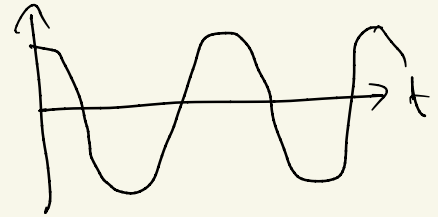
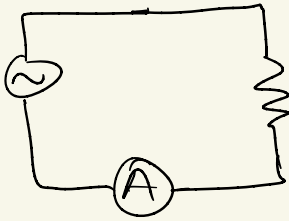
Note 3:

Nonlinear elements

(e.g. Diodes)



$\xi \cos \omega t$



PN junction

p: p-doped semiconductor

n: n-doped semiconductor

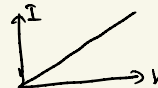
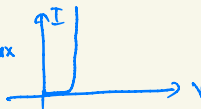


Zener diodes

Linear behavior = Ohmic behavior

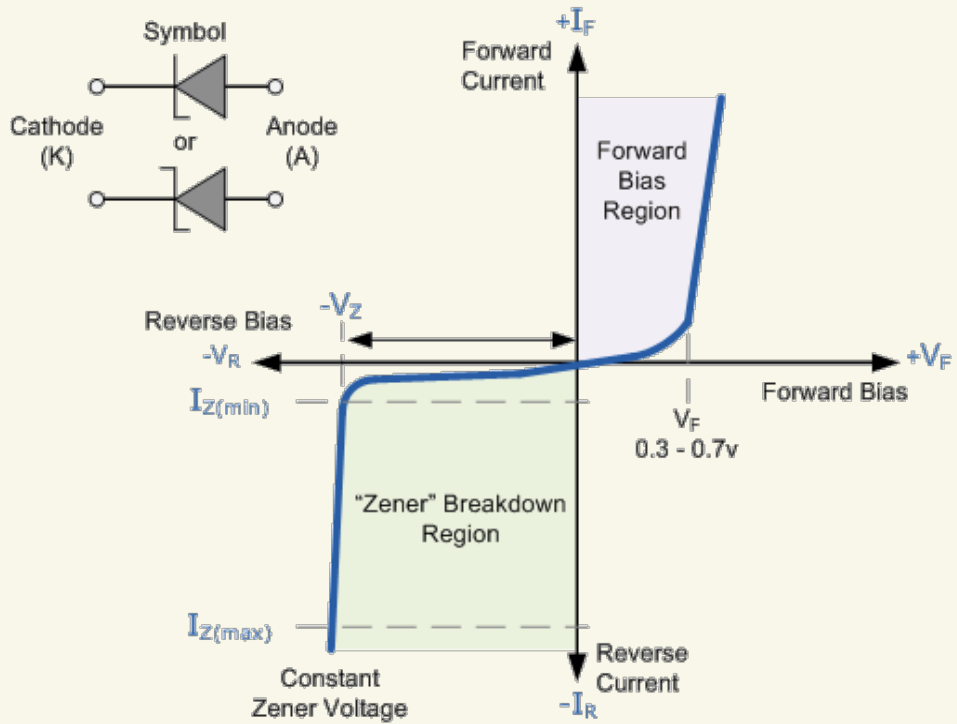
$$V = IR$$

В диодах



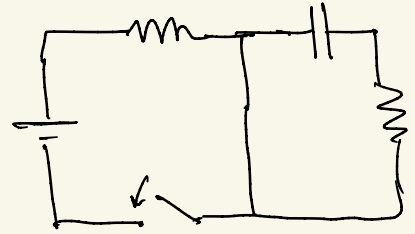
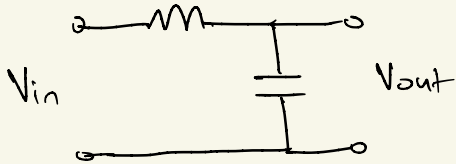
ВАХ (вольт-амперная характеристика)

Case: Zener diodes $\rightarrow$ 



Note 4: Filters (high-pass, low pass)

Heart of Electronics !!!



Remember: voltage divider

$$V_{out} = I X_C$$

$$V_{in} = I R + I X_C$$

$$I = I$$

$$\frac{V_{out}}{X_C} = \frac{V_{in}}{R + X_C}$$

$$V_{out} = \frac{X_C}{R + X_C} V_{in}$$

Помните!, $X_C = \frac{1}{\omega C}$, сократим

$$V_{out} = \frac{\frac{1}{\omega C}}{R + \frac{1}{\omega C}} V_{in}$$

Здесь ω - частота колебаний $V_{in} = E_0 \sin(\omega t + \phi)$

Observation:

$$V_{out} = \frac{\frac{1}{\omega C}}{R + \frac{1}{\omega C}} V_{in} = \frac{1}{\omega RC + 1} V_{in}$$

at $\omega \rightarrow 0$
(low freq)

$$V_{out} = \lim_{\omega \rightarrow 0} \frac{1}{\omega RC + 1} V_{in} = \frac{1}{0+1} V_{in} = V_{in}$$

(max)

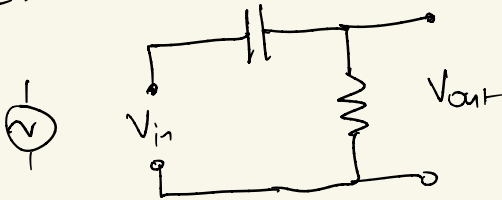
at $\omega \rightarrow \infty$
(high freq)

$$\begin{aligned} V_{out} &= \lim_{\omega \rightarrow \infty} \frac{1}{1 + R\omega C} V_{in} \\ &= \frac{1}{1 + \infty} V_{in} \\ &= 0. \end{aligned}$$

(min)

low-pass filter = passes low freq signals

Ex2:



Similarly, $\sim$

$$V_{out} = \frac{R}{(\omega C)^{-1} + R} V_{in}$$

• at $\omega \rightarrow 0$ (low freq)

$$V_{out} = \frac{R}{\left(\frac{1}{\omega C}\right) + R} V_{in}$$

$$= \frac{R}{\frac{1}{0} + R} V_{in} = \frac{R}{\infty} V_{in}$$

$$= 0.$$

high-pass filter

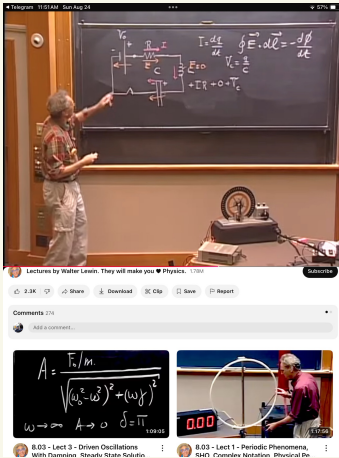
• at $\omega \rightarrow \infty$ (high freq).

$$V_{out} = \lim_{\omega \rightarrow \infty} \frac{R}{\frac{1}{\omega C} + R} V_{in} = \frac{R}{\frac{1}{\infty} + R} V_{in} = V_{in}$$

HW Ch 31

Questions: 1, 5, 7, 8

Problems: 8, 9, 13, 15, 17, 20, 24, 25, 26, 27, 28



RLC damped
watch Walter Lewin lecture

https://www.youtube.com/watch?v=pnfZjGlyKVw&ab_channel=LecturesbyWalterLewin.Theywillmakeyou%E2%99%A5Physics

31, 33, 36, 38, 42, 43, 44, 45, 47

48, 49, 54, 58, 59, 64, 65

↓
typical olympiad problem

Note: remember that your ultimate target for EM and this chapter is completing Kevin Zhou Syllabus along with Purcell & Cabretnoo!!

Deadline: Aug 31 (Sun) noon time 12:00 PM.

Note 5: RMS values in time-varying sources

$$\begin{aligned} Z &= \sqrt{R^2 + (X_L - X_C)^2} \\ &= \sqrt{(200 \Omega)^2 + (86.7 \Omega - 177 \Omega)^2} \\ &= 219 \Omega. \end{aligned}$$



Additional examples, video, and practice available at WileyPLUS

The negative phase constant is consistent with the fact that the load is mainly capacitive; that is, $X_C > X_L$. In the common mnemonic for driven series *RLC* circuits, this circuit is an *ICE* circuit—the current *leads* the driving emf.

31-10 Power in Alternating-Current Circuits

In the *RLC* circuit of Fig. 31-7, the source of energy is the alternating-current generator. Some of the energy that it provides is stored in the electric field in the capacitor, some is stored in the magnetic field in the inductor, and some is dissipated as thermal energy in the resistor. In steady-state operation, the average stored energy remains constant. The net transfer of energy is thus from the generator to the resistor, where energy is dissipated.

The instantaneous rate at which energy is dissipated in the resistor can be written, with the help of Eqs. 26-27 and 31-29, as

$$P = i^2 R = [I \sin(\omega_d t - \phi)]^2 R = I^2 R \sin^2(\omega_d t - \phi). \quad (31-68)$$

The *average* rate at which energy is dissipated in the resistor, however, is the average of Eq. 31-68 over time. Over one complete cycle, the average value of $\sin \theta$, where θ is any variable, is zero (Fig. 31-17a) but the average value of $\sin^2 \theta$ is $\frac{1}{2}$ (Fig. 31-17b). (Note in Fig. 31-17b how the shaded areas under the curve but above the horizontal line marked $+\frac{1}{2}$ exactly fill in the unshaded spaces below that line.) Thus, we can write, from Eq. 31-68,

$$P_{\text{avg}} = \frac{I^2 R}{2} = \left(\frac{I}{\sqrt{2}} \right)^2 R. \quad (31-69)$$

The quantity $I/\sqrt{2}$ is called the **root-mean-square**, or **rms**, value of the current i :

$$I_{\text{rms}} = \frac{I}{\sqrt{2}} \quad (\text{rms current}). \quad (31-70)$$

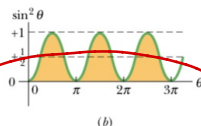
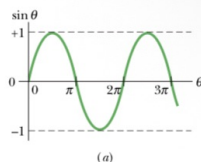


Fig. 31-17 (a) A plot of $\sin \theta$ versus θ . The average value over one cycle is zero. (b) A plot of $\sin^2 \theta$ versus θ . The average value over one cycle is $\frac{1}{2}$.

Logic: $V_{\text{in}} = \mathcal{E}_0 \sin(\omega_d t)$, $\langle \sin \omega_d t \rangle = 0$
 $\langle \sin^2 \omega_d t \rangle = \frac{1}{2}$ Power
 BUT $\leftarrow$ if you average $\sin(\omega_d t) = 0$ voltages & $\rightarrow$
 BUT we know that $V_{\text{in}} \neq 0$, so we need to find a smarter way to estimate
 so you square it and take its avg instd.

$$\langle f(t) \rangle = \frac{1}{T} \int_0^T f(t) dt$$

$$\langle \sin(\omega t) \rangle = \frac{1}{T} \int_0^T \sin \omega t dt = 0,$$

$$\langle \cos(\omega t) \rangle = \frac{1}{T} \int_0^T \cos \omega t dt = 0, \quad T = 2\pi\omega^{-1}$$

$$\begin{aligned} \langle \sin^2(\omega t) \rangle &= \frac{1}{T} \int_0^T \sin^2 \omega t dt \\ &= \frac{1}{T} \left(\frac{t}{2} - \frac{\sin(2\omega t)}{4\omega} \right) \\ &= \frac{1}{2}. \end{aligned}$$